Skills Summary

Sign Errors in Parabola Transformations

Use this reference while completing your worksheet or when studying.

Vertex form: $y = a(x-h)^2 + k$ vertex (h, k), axis of symmetry $x = h$.

The one question that settles every sign: *what value of x makes the squared factor zero?* That value **is** h. Read h that way and you never have to remember a direction rule.

a multiplies **outputs**: $|a| > 1$ stretches, $0 < |a| < 1$ compresses, and $a < 0$ flips the parabola over its own vertex. a never moves the vertex.

Skill 1 — Reading the vertex out of the form

Rule: the horizontal shift goes *opposite* the printed sign, because the form subtracts h.

$(x-5) \Rightarrow h = 5$: right 5

$(x+3) \Rightarrow h = -3$: left 3

The vertical part has no such twist: $+k$ is up, $-k$ is down.

Example: Find the minimum value of $y = (x-2)^2 + 6$.

Step 1. The minimum of an upward parabola happens at the vertex, so I need h — the x that makes the squared factor zero.

Step 2. $x - 2 = 0$ gives $x = 2$ (right 2, not left). Then $y = (2-2)^2 + 6 = 0 + 6$. $\Rightarrow$ minimum value $=$ **6** at the point $(2, 6)$

Try it: Find the minimum value of $y = (x-7)^2 + 1$.

check: **1**

Skill 2 — Writing the equation from a description

Build it in this order: reflection and stretch into a, horizontal shift inside the parentheses (*opposite* sign), vertical shift outside (same sign).

right p: $(x-p)$ left p: $(x+p)$ up q: $+q$ down q: $-q$ reflect over the x-axis: $a < 0$

Example: Reflect $y = x^2$ over the x-axis, shift left 3 and up 10. What is the value at $x = -3$?

Step 1. Each described move maps to one slot of $a(x-h)^2 + k$, so I write the equation first and only then substitute.

Step 2. Reflection gives $a = -1$, left 3 gives $(x+3)$, up 10 adds 10: $y = -(x+3)^2 + 10$. Then $y(-3) = -(-3+3)^2 + 10 = 0 + 10$. $\Rightarrow$ **10**

Try it: Stretch $y = x^2$ vertically by 2, shift left 5 and down 6. What is the value at $x = -5$?

check: **−6**

Skill 3 — Finding and fixing a sign error

How to audit someone's work in three checks:

1. Does their h make the squared factor zero? If not, the shift direction is flipped.

2. Is their vertex value the largest or smallest output? A wrong h still gives a real point — it is just not the vertex.

3. Did the sign of a survive? Losing it turns a maximum into a minimum.

Example: A student says the vertex of $y = (x + 8)^2 - 2$ is at $x = 8$ and the minimum is 254. Fix it.

Step 1. Check 1 first: does their h zero the squared factor? $8 + 8 = 16 \neq 0$, so their h is the culprit.

Step 2. $x + 8 = 0$ gives $x = -8$, so $y = (-8 + 8)^2 - 2 = 0 - 2$. Their 254 is $y(8)$ — a genuine point on the curve, just not the lowest. $\Rightarrow$ true minimum $=$ $\mathbf{-2}$ at $(-8, -2)$

Try it: A student reports the vertex of $y = -(x - 9)^2 + 4$ as $x = -9$, with maximum -320. What is the real maximum?

check: **4**

(!) Watch out: A wrong h almost never produces an obviously silly number — it produces a real point on the parabola, which is why these errors survive. Always test the candidate vertex by checking that the squared factor is zero there. And remember that a negative a reflects the graph without moving it: the shifts stay exactly where h and k put them.